

XIE Jintao

Hong Kong SAR | u3593115@connect.hku.hk | github.com/jintaoxie02

PROFILE

Year 3 BEng Data Science and Engineering student (Minor: Digital Health) seeking an actuarial internship. Passed SOA Exam P and IIQE Papers I & III. Strong analytical foundation with expertise in Python, R, statistics, and data modeling. Proven ability to leverage programming and machine learning to solve complex problems and build data-driven solutions.

EDUCATION

BEng in Data Science and Engineering, Minor in Digital Health

2023–2027

The University of Hong Kong

ACTUARIAL EXAMS & CERTIFICATIONS

Society of Actuaries Exam P

IIQE Paper I: Principles & Practice of Insurance

IIQE Paper III: Long Term Insurance

(Cred

— Voc

— Voc

TECHNICAL SKILLS

Languages Python, R, SQL, TypeScript

Data / Modeling numpy, pandas, matplotlib, scikit-learn, statistical modeling, time series analysis

Topics Data analysis, machine learning, predictive modeling, probability

Other Tech Next.js, LangChain, Arduino, ESP32

EXPERIENCE

Programmer (Freelance) — AIGreen

Jul 2025 – Present

Hong Kong SAR · Hybrid

- Developed control software with hardware interfacing, demonstrating strong logical problem-solving and attention to detail.
- Implemented iterative prototype testing and calibration, ensuring high accuracy and reliability of data collection.
- Delivered rapid updates and coordinated requirements effectively in a fast-paced, independent working environment.

PROJECTS

EEG-Based Computational Psychiatry & Accessibility

Jan 2026 – Present

HKU Innovation Wing, Faculty of Engineering

Statistical Modeling | Time Series Analysis | Python | Health Data

- Analyzing high-dimensional EEG time-series data to develop predictive models for mental health conditions, demonstrating strong capability in handling complex biometric datasets.
- Applying advanced statistical techniques to extract actionable insights, highly relevant to health insurance and mortality/morbidity risk assessment.

Mental Health Support App

Aug 2025

The University of Hong Kong

Next.js | TypeScript | LangChain | Data Privacy

- Built a responsive web app integrating a conversation pipeline to support consistent multi-turn interactions.
- Designed data flows and messaging with strict confidentiality-first principles, aligning with data security standards.

InnoGrow: Smart Solutions for a Sustainable Future

Dec 2024 – Jul 2025

The University of Hong Kong

Python | Deep Learning | Computer Vision | Data Analysis

- Contributed to vision-based perception models, applying statistical and machine learning techniques to real-world data.

- Collaborated with domain experts to refine requirements, evaluate model performance, and optimize outcomes.
- InsurPrime: Can You Guess the Insurance Premium?** Jan 2026
- Challenge Data / ENS Paris — Organized by Crédit Agricole Assurances
- Python | Machine Learning | Predictive Modeling | Insurance Analytics
- Ranked 42nd (top 11.05%) globally in a supervised ML competition predicting insurance premiums from real-world policyholder data.
 - Built and optimized predictive models for premium estimation, applying regression techniques and feature engineering to actuarial pricing data.

- Memory-OS: Hierarchical Memory System for AI Agents** Jul 2026
- github.com/jintaoxie02/memory-os
- Python | LLM Integration | SQLite + ChromaDB | LangChain | MIT Licensed
- Designed and built a three-layer memory hierarchy (hot/warm/cold) for LLM agents, inspired by cognitive science principles of memory consolidation and retrieval.
 - Implemented LLM-powered extraction and merging engines with a state-machine evolution system (new → confirmed → modified → archived), plus hybrid retrieval scoring combining semantic similarity, recency decay, importance, and state bonus.

AWARDS

- HKU Engineering Recognition Ceremony (Undergraduate Students 2024-25)
- Winner, HKAE Pitch Competition 2024-2025

LANGUAGES

Cantonese (Native) · Mandarin (Native) · English (Fluent)

ADDITIONAL INFORMATION

Open to on-site, hybrid, or remote roles.